

Autonomous Expert Systems and Data Exploration

30 Student Project Proposals

Data Processing, REST API Ingestion, Spark/Python, AWS Educate, Analytics and AI/ML

Course Context

The projects are prepared for the course Autonomous Expert Systems and Data Exploration. The main concept of the course is data processing: collecting data from an external API, storing raw data, transforming and validating datasets, building analytical outputs, and presenting conclusions. AI/ML and expert-system components may be used, but they should support the data-processing workflow rather than replace it.

Shared Data Source

All projects should use the shared Weather REST API as the primary source of data. Students should not invent the main dataset manually. They may create small configuration files, labels, rules, dictionaries, or test cases, but the core observations should be collected from the API.

Team Work Rules

Projects are designed for teams of 2 students. A team of 3 students is possible only with the lecturer's permission. Larger teams should deliver a broader implementation scope, for example more stations, more complete automation, stronger evaluation, or a richer dashboard.

Recommended Minimal Architecture

REST API collector -> raw storage -> validation/cleaning -> transformation/feature engineering -> analytics/ML/expert rules -> report/dashboard/presentation.

Shared Weather REST API

The following API usage section is the only part reused from the Intelligent Measurement Systems project description. In this document, the API is treated only as a common data source for data-processing projects.

Base URL	https://e6uw49pbah.execute-api.us-east-1.amazonaws.com/dev
Authentication header	Authorization: Bearer STUDENT_TOKEN_2026
Latest measurement	GET /weather/latest?station_id=GDN_01
Batch measurements	GET /weather/batch?station_id=GDN_01&limit=100
Available stations	GET /weather/stations

Example API request:

```
curl -X GET
  "https://e6uw49pbah.execute-api.us-east-1.amazonaws.com/dev/weather/latest?station_id=GDN_01" \
  -H "Authorization: Bearer STUDENT_TOKEN_2026"
```

Example response fields include timestamp, station_id, temperature, humidity, pressure, wind_speed, wind_direction, rain_mm, and cloud_cover.

Common Project Requirements

- Use the shared Weather REST API as the main data source.
- Persist raw data before cleaning or aggregation.
- Clearly separate data ingestion, processing, analysis, and presentation steps.
- Use Python, pandas, PySpark/Spark SQL, AWS Glue/EMR, Athena, DuckDB, or similar tools where appropriate.

- Include at least one meaningful analytical result: model, metric, score, classification, dashboard, or report.
- Document assumptions, limitations, and possible improvements.

Suggested Minimal Deliverables per Team

- README with project goal, architecture, and execution steps.
- Source code or notebook for ingestion and analysis.
- Storage structure: local files, S3 folders, database tables, or lakehouse-style layers.
- Final report with charts, tables, metrics, and conclusions.
- Short presentation explaining the data pipeline, analytical method, and results.

Project Overview

No.	Project Title	Main Data-Processing Focus
1	Temperature Anomaly Detection	API batch/latest ingestion, timestamp normalization, rolling windows, anomaly flags, curated anomaly table.
2	Temperature Forecasting	collect historical batches, build lag features, split train/test data, persist forecasts.
3	Weather Comfort Index	process API measurements into hourly comfort scores by station and activity.
4	Weather Alerting System	periodic API collection, threshold evaluation, alert log, optional notification workflow.
5	Weather Type Classification	transform raw API records into labelled examples and reusable classification output.
6	Weather Data Quality Monitoring	raw-to-clean processing, validation rules, quarantine invalid records, quality score.
7	Multi-Station Weather Comparison	collect data for several station IDs, align time windows, join station datasets, build comparison features.
8	Mini Lakehouse for Weather Data	organize API data into Bronze/Silver/Gold layers using local files or S3.
9	Weather Event Timeline and Automatic Daily Report	daily ingestion, hourly aggregations, event extraction, report table generation.
10	Root-Cause Analysis for Weather Anomalies	join anomaly flags with feature windows before and after each event.
11	Weather Pattern Clustering	build feature vectors from API data, scale variables, persist cluster assignments.
12	Rain Risk Scoring and Short-Term Prediction	create rolling features for humidity, cloud cover, pressure, wind, and previous rain.
13	Wind Risk Analysis for Outdoor Activities	aggregate wind speed by hour/station and combine it with weather context.
14	Pressure Drop Detection and Storm-Risk Expert System	calculate pressure deltas and combine them with wind, rain, and cloud cover.
15	Data Drift Monitoring Between Stations	create rolling station profiles and compare them to previous periods or other stations.
16	Feature Engineering Pipeline for Weather ML	process API data into a feature table with time, lag, rolling, and categorical features.
17	Natural-Language Weather Data Explorer	curate API data, define supported question templates, validate generated queries.

18	Weather Knowledge Base and RAG-Style Assistant	extract facts from processed API data and retrieve relevant rule/context snippets.
19	Near-Real-Time Weather Dashboard Pipeline	scheduled API collection, latest-state table, history storage, dashboard dataset.
20	Sensor Reliability Ranking and Station Trust Score	compute quality and consistency metrics for each station from API records.
21	Weather Similarity Search Across Days	aggregate API records into daily/hourly feature vectors and compare them.
22	Microclimate Discovery for Local Areas	collect multi-station data and compare station profiles over aligned time periods.
23	Automated Data Cleaning Recommendation System	classify quality issues and generate cleaning actions for Silver-layer processing.
24	Cost-Aware Weather Data Lake on AWS	store API data as JSON and Parquet, compare size and query behavior.
25	Weather-Based Activity Recommendation Engine	create scoring inputs from API data and rank activities per station/time.
26	Explainable ML for Weather Type Prediction	prepare labelled/rule-labelled API data, train a model, evaluate feature influence.
27	Batch vs Near-Real-Time Processing Comparison	implement or design two processing paths using the same API data and metrics.
28	Autonomous End-to-End Weather Analytics Agent	connect approved query tools to processed API data and log all actions.
29	Metadata, Lineage, and Data Catalog for Weather Processing	capture metadata for API ingestion, transformations, tables, columns, and reports.
30	Self-Optimizing Weather Data Pipeline	collect API data, measure runtime/quality/volume metrics, recommend tuning actions.

Project 1 - Temperature Anomaly Detection

Goal

Detect unusual temperature behavior in a stream or batch of weather measurements.

Data Processing Scope

API batch/latest ingestion, timestamp normalization, rolling windows, anomaly flags, curated anomaly table.

Suggested Analytical Methods

- moving average
- rolling standard deviation
- Z-score
- IQR
- rule-based detection

Expected Output

- anomaly table
- chart with normal/anomalous points
- method explanation
- conclusions

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 2 - Temperature Forecasting

Goal

Predict future temperature values using historical weather measurements.

Data Processing Scope

collect historical batches, build lag features, split train/test data, persist forecasts.

Suggested Analytical Methods

- linear regression
- random forest
- moving-average baseline
- time-based features

Expected Output

- forecast dataset
- MAE/RMSE/MAPE metrics
- predicted vs actual chart

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 3 - Weather Comfort Index

Goal

Calculate and visualize a comfort score for outdoor activities.

Data Processing Scope

process API measurements into hourly comfort scores by station and activity.

Suggested Analytical Methods

- rule-based scoring
- weighted variables
- activity-specific thresholds

Expected Output

- comfort score
- comfort class
- dashboard/report
- activity recommendation

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 4 - Weather Alerting System

Goal

Build an alerting pipeline for dangerous or important weather conditions.

Data Processing Scope

periodic API collection, threshold evaluation, alert log, optional notification workflow.

Suggested Analytical Methods

- rule engine
- trend-based rules
- false-positive reduction

Expected Output

- alert history
- triggered rules
- optional SNS/email simulation
- operational notes

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 5 - Weather Type Classification

Goal

Classify weather conditions into useful categories such as rainy, windy, cloudy, warm, cold, unstable, or storm-risk.

Data Processing Scope

transform raw API records into labelled examples and reusable classification output.

Suggested Analytical Methods

- rule-based classification
- supervised model
- confusion matrix if labels are available

Expected Output

- classified dataset
- visual summary
- comparison of rules vs ML

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 6 - Weather Data Quality Monitoring

Goal

Evaluate completeness, validity, and consistency of incoming weather measurements.

Data Processing Scope

raw-to-clean processing, validation rules, quarantine invalid records, quality score.

Suggested Analytical Methods

- missing/duplicate checks
- range checks
- timestamp validation
- consistency checks

Expected Output

- data quality report
- invalid records

Autonomous Expert Systems and Data Exploration

- quality score
- cleaning recommendations

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 7 - Multi-Station Weather Comparison

Goal

Compare weather behavior between multiple stations.

Data Processing Scope

collect data for several station IDs, align time windows, join station datasets, build comparison features.

Suggested Analytical Methods

- station-level aggregations
- correlation
- ranking
- visualization

Expected Output

- multi-station dataset
- station comparison charts
- microclimate hypotheses

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 8 - Mini Lakehouse for Weather Data

Goal

Build a small data lakehouse-inspired pipeline with raw, cleaned, and analytical layers.

Data Processing Scope

organize API data into Bronze/Silver/Gold layers using local files or S3.

Suggested Analytical Methods

- schema design
- validation
- transformations
- aggregations
- SQL access

Expected Output

- documented data layers
- curated tables
- analytical queries
- pipeline documentation

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 9 - Weather Event Timeline and Automatic Daily Report

Goal

Generate a daily timeline of important weather events and a readable analytical summary.

Data Processing Scope

daily ingestion, hourly aggregations, event extraction, report table generation.

Suggested Analytical Methods

- Spark/Python windows
- rule-based event detection
- optional LLM summary

Expected Output

- daily event timeline
- key-event table
- generated report
- limitations

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 10 - Root-Cause Analysis for Weather Anomalies

Goal

Explain likely reasons behind detected anomalies using related variables.

Data Processing Scope

join anomaly flags with feature windows before and after each event.

Suggested Analytical Methods

- lag features
- correlation around anomaly windows
- decision-tree explanations

Expected Output

- root-cause candidate table
- timeline around anomaly
- expert-style reasoning

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 11 - Weather Pattern Clustering

Goal

Discover groups of similar weather situations.

Data Processing Scope

build feature vectors from API data, scale variables, persist cluster assignments.

Suggested Analytical Methods

- K-means
- PCA optional
- cluster profiling

Expected Output

- clustered dataset
- cluster descriptions
- charts
- recommended cluster names

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 12 - Rain Risk Scoring and Short-Term Prediction

Goal

Estimate near-future rain risk from recent measurements.

Data Processing Scope

create rolling features for humidity, cloud cover, pressure, wind, and previous rain.

Suggested Analytical Methods

- binary classification
- baseline rules
- precision/recall evaluation

Expected Output

- rain-risk score 0-100
- model/rules
- evaluation results

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 13 - Wind Risk Analysis for Outdoor Activities

Goal

Classify wind conditions and estimate risk for cycling, running, walking, or field work.

Data Processing Scope

aggregate wind speed by hour/station and combine it with weather context.

Suggested Analytical Methods

- risk scoring
- sudden-change proxy
- activity thresholds

Expected Output

- wind-risk score
- activity table
- hourly heatmap

- threshold explanation

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 14 - Pressure Drop Detection and Storm-Risk Expert System

Goal

Monitor pressure changes and estimate unstable-weather risk.

Data Processing Scope

calculate pressure deltas and combine them with wind, rain, and cloud cover.

Suggested Analytical Methods

- rolling pressure trends
- expert rules
- risk classification

Expected Output

- storm-risk categories
- alert log
- high-risk case study

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 15 - Data Drift Monitoring Between Stations

Goal

Detect changes in statistical behavior over time or across stations.

Data Processing Scope

create rolling station profiles and compare them to previous periods or other stations.

Suggested Analytical Methods

- distribution comparison
- rolling mean/variance
- PSI optional

Expected Output

- drift score
- drift alerts
- distribution charts
- interpretation

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 16 - Feature Engineering Pipeline for Weather ML

Goal

Create a reusable feature pipeline for multiple analytical and ML tasks.

Data Processing Scope

process API data into a feature table with time, lag, rolling, and categorical features.

Suggested Analytical Methods

- feature generation
- feature quality checks
- leakage prevention

Expected Output

- feature table
- feature dictionary
- example model using features

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 17 - Natural-Language Weather Data Explorer

Goal

Create a small assistant that maps controlled natural-language questions to SQL or Spark queries.

Data Processing Scope

curate API data, define supported question templates, validate generated queries.

Suggested Analytical Methods

- template-based query mapping
- SQL validation
- result summarization

Expected Output

- Q&A interface
- supported questions
- generated queries
- result explanations

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 18 - Weather Knowledge Base and RAG-Style Assistant

Goal

Combine weather measurements with a small knowledge base of rules and definitions.

Data Processing Scope

extract facts from processed API data and retrieve relevant rule/context snippets.

Suggested Analytical Methods

- chunking
- keyword/vector retrieval
- grounded answer generation

Expected Output

- knowledge base
- retrieval logic
- example answers
- grounding evaluation

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 19 - Near-Real-Time Weather Dashboard Pipeline

Goal

Build a periodically refreshed dashboard pipeline.

Data Processing Scope

scheduled API collection, latest-state table, history storage, dashboard dataset.

Suggested Analytical Methods

- incremental ingestion
- deduplication
- hourly aggregation
- latency measurement

Expected Output

- working schedule
- dashboard/report
- latest measurements
- operational documentation

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 20 - Sensor Reliability Ranking and Station Trust Score

Goal

Rank stations using data completeness, consistency, and anomaly frequency.

Data Processing Scope

compute quality and consistency metrics for each station from API records.

Suggested Analytical Methods

- quality checks
- weighted scoring
- cross-station consistency

Expected Output

- station trust score

Autonomous Expert Systems and Data Exploration

- station ranking
- issue breakdown
- monitoring recommendation

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 21 - Weather Similarity Search Across Days

Goal

Find historical periods most similar to a selected day or hour.

Data Processing Scope

aggregate API records into daily/hourly feature vectors and compare them.

Suggested Analytical Methods

- cosine or Euclidean distance
- normalization
- nearest-neighbor search

Expected Output

- similarity notebook
- ranked similar periods
- comparison charts

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 22 - Microclimate Discovery for Local Areas

Goal

Identify local differences between stations and explain possible microclimate patterns.

Data Processing Scope

collect multi-station data and compare station profiles over aligned time periods.

Suggested Analytical Methods

- correlation matrix
- profile comparison
- clustering optional

Expected Output

- microclimate report
- station profiles
- charts
- hypotheses

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 23 - Automated Data Cleaning Recommendation System

Goal

Recommend how to handle detected data quality problems.

Data Processing Scope

classify quality issues and generate cleaning actions for Silver-layer processing.

Suggested Analytical Methods

- repair strategies
- imputation
- quarantine
- audit trail

Expected Output

- cleaning recommendation table
- cleaned dataset
- quarantined records

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 24 - Cost-Aware Weather Data Lake on AWS

Goal

Design and measure efficient storage and query patterns for weather data.

Data Processing Scope

store API data as JSON and Parquet, compare size and query behavior.

Suggested Analytical Methods

- partitioning
- compaction
- format comparison
- query-cost reasoning

Expected Output

- S3 layout recommendation
- JSON vs Parquet comparison
- cost/performance conclusions

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 25 - Weather-Based Activity Recommendation Engine

Goal

Recommend activities based on current and recent weather conditions.

Data Processing Scope

create scoring inputs from API data and rank activities per station/time.

Suggested Analytical Methods

- multi-criteria scoring
- activity rules
- synthetic labels optional

Expected Output

- ranked recommendations
- explanation per activity
- scenario tests

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 26 - Explainable ML for Weather Type Prediction

Goal

Build a weather classifier and explain model decisions.

Data Processing Scope

prepare labelled/rule-labelled API data, train a model, evaluate feature influence.

Suggested Analytical Methods

- classification
- feature importance
- decision-tree interpretation
- error analysis

Expected Output

- trained classifier
- prediction explanations
- metrics
- rule-vs-ML comparison

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 27 - Batch vs Near-Real-Time Processing Comparison

Goal

Compare batch and near-real-time designs for the same analytical task.

Data Processing Scope

implement or design two processing paths using the same API data and metrics.

Suggested Analytical Methods

- latency comparison
- duplicate handling
- state management
- complexity analysis

Expected Output

- two designs

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- measured/estimated latency
- pros/cons table
- recommendation

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 28 - Autonomous End-to-End Weather Analytics Agent

Goal

Design a controlled agent workflow that plans, runs, validates, and summarizes weather analysis.

Data Processing Scope

connect approved query tools to processed API data and log all actions.

Suggested Analytical Methods

- task planning
- safe query execution
- result validation
- automatic summary

Expected Output

- agent workflow diagram
- allowed tasks
- execution log
- analytical summary

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 29 - Metadata, Lineage, and Data Catalog for Weather Processing

Goal

Track where data comes from, how it changes, and which outputs depend on each processing step.

Data Processing Scope

capture metadata for API ingestion, transformations, tables, columns, and reports.

Suggested Analytical Methods

- lineage mapping
- schema registry
- metadata tables
- documentation automation

Expected Output

- data catalog
- lineage diagram/table
- schema documentation
- impact-analysis examples

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Project 30 - Self-Optimizing Weather Data Pipeline

Goal

Build a pipeline that monitors its own processing quality and suggests operational improvements.

Data Processing Scope

collect API data, measure runtime/quality/volume metrics, recommend tuning actions.

Suggested Analytical Methods

- pipeline observability
- rule-based optimization
- retry logic
- threshold tuning

Expected Output

- pipeline health report
- optimization recommendations
- example before/after improvement

Reference Architecture

Weather REST API -> raw storage -> processing/validation -> curated dataset -> analytical output -> report/dashboard.

Common Evaluation Criteria

- Correct and repeatable data ingestion from the REST API, including token usage.
- Clear data-processing design with raw and processed data separated.
- Readable code, meaningful transformations, and documented execution steps.
- Quality of analysis, metrics, charts, and final conclusions.
- Ability to explain limitations, assumptions, and future improvements.
- End-to-end completeness within the constraints of student AWS Educate or local environments.